A

Report on

PRACTICAL SECURITY ASSESSMENT (TASK 7)

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1. VULNERABILITY SCANNING LAB

1.1 Objective

The objective of this lab is to identify vulnerabilities present in a system using automated scanning tools. This helps in detecting security weaknesses before they can be exploited by attackers.

1.2 Tools Used

- Nmap – for network scanning
- Nikto – for web server vulnerability scanning
- Kali Linux – penetration testing environment

1.3 Procedure

- Set up the lab environment using Kali Linux and a target machine.
- Identify the IP address of the target system.
- Perform network scanning using Nmap.
- Conduct web vulnerability scanning using Nikto.
- Analyze the results to identify vulnerabilities.

1.4 Commands Used

- nmap -sV <target_ip> → Service version detection
- nmap -A <target_ip> → Aggressive scan
- nikto -h http://<target_ip> → Web vulnerability scan

1.5 Objectives

- Multiple open ports were detected such as 21, 22, and 80.
- Services like FTP, SSH, and HTTP were running.
- Some services were outdated and vulnerable.
- Nikto detected misconfigurations and security issues.

1.6 Result

The scanning process successfully identified vulnerabilities in the target system. These vulnerabilities can be further tested during penetration testing.

2. RECONNAISSANCE PRACTICE

2.1 Objective

The objective of reconnaissance is to gather information about the target system before launching an attack.

2.2 Tools Used

- Whois
- Nslookup
- Shodan
- Nmap

2.3 Procedure

- Collect basic information such as IP address and domain name.
- Use Whois to gather domain registration details.
- Perform DNS lookup using Nslookup.
- Use Shodan to identify exposed devices.
- Perform initial scanning using Nmap.

2.4 Observations

- Domain and IP information were successfully collected.
- DNS records provided useful insights.
- Some services were publicly exposed.

2.5 Result

Reconnaissance provided valuable information about the target, which is essential for further testing phases.

3. EXPLOITATION LAB

3.1 Objective

The objective of this lab is to exploit identified vulnerabilities and gain access to the target system.

3.2 Tools Used

- Metasploit Framework
- Kali Linux

3.3 Procedure

- Launch Metasploit framework.
- Search for available exploits.
- Select a suitable exploit based on the vulnerability.
- Configure exploit parameters.
- Execute the exploit.

3.4 Commands Used

- msfconsole → Start Metasploit
- search <exploit> → Search exploits
- use <exploit_name> → Select exploit
- set RHOST <target_ip> → Set target
- exploit → Execute attack

3.5 Observations

- Exploits were successfully executed on vulnerable services.
- Access to the target system was achieved in some cases.
- Some exploits failed due to patched vulnerabilities.

3.6 Result

The exploitation phase demonstrated how vulnerabilities can be used to gain unauthorized access to systems.

4. POST-EXPLOITATION PRACTICE

4.1 Objective

The objective is to maintain access and extract useful information after successful exploitation.

4.2 Tools Used

- Metasploit
- Meterpreter

4.3 Procedure

- Gain access to the system using exploitation techniques.
- Use Meterpreter session for post-exploitation.
- Collect system information.
- Maintain access if required.

4.4 Evidence Collection

- System information (OS, user details)
- Network configuration
- Sensitive files and credentials
- Screenshots (if applicable)

4.5 Result

Post-exploitation provided deeper insights into system security and demonstrated the impact of vulnerabilities.

5. CAPSTONE PROJECT: FULL VAPT CYCLE

5.1 Objective

The objective of this project is to perform a complete VAPT cycle including scanning, reconnaissance, exploitation, and reporting.

5.2 Tools Used

- Nmap
- Nikto
- Metasploit
- Shodan
- Kali Linux

5.3 Procedure

- Perform reconnaissance to gather information.
- Conduct vulnerability scanning.
- Identify security weaknesses.
- Exploit vulnerabilities.
- Perform post-exploitation analysis.
- Document findings and prepare report.

5.4 Commands Used

- nmap -A <target_ip>
- nikto -h http://<target_ip>
- msfconsole
- exploit

5.5 Observations

- Multiple vulnerabilities were identified across different services.
- Some exploits successfully provided system access.
- Security misconfigurations were observed.

5.6 Result

The full VAPT cycle successfully demonstrated the process of identifying and exploiting vulnerabilities. The system was found to have several security weaknesses that need to be addressed.